

Cooling Systems & Power Management

1. Measure your CPU temperature at idle and under load using a tool like HWMonitor or Core Temp, then record the readings before and after applying new thermal paste and reinstalling your CPU cooler.

-> You can use the following **assignment-ready table** to record your actual CPU temperature readings. Replace the example values with the temperatures you measure.

Condition	Before Thermal Paste	After Thermal Paste
Idle	48°C	38°C
Under Load	82°C	68°C

Procedure

1. Open **HWMonitor** or **Core Temp** and record the CPU temperature after the PC has been idle for about 10 minutes.
2. Run a CPU-intensive task for about **5–10 minutes** and record the highest temperature reached.
3. Shut down the PC, unplug it, and safely remove the CPU cooler.
4. Clean the old thermal paste and apply a **small pea-sized amount** of new thermal paste.
5. Reinstall the CPU cooler evenly and securely.
6. Repeat the idle and load tests under similar conditions.

Short Observation

The example results show a decrease from **48°C to 38°C at idle** and from **82°C to 68°C under load** after replacing the thermal paste and reinstalling the cooler. This indicates improved heat transfer from the CPU to the cooler. **Use your actual measured readings in the final report rather than these example values.**

2. Write a step-by-step guide (with photos or diagrams) showing how to correctly apply thermal paste and mount a CPU cooler for optimal heat transfer.

Hint:

Include tips on how much paste to use, spreading methods, and common mistakes to avoid.

-> Here is an assignment-ready guide you can use. I can also create a labeled diagram for the steps if you want.

Step-by-Step: Applying Thermal Paste and Mounting a CPU Cooler

1. Turn off and prepare the PC

- Shut down the computer and unplug the power cable.
- Place the PC on a clean, stable surface.
- Ground yourself or use an anti-static wrist strap.

2. Clean the CPU and cooler

- Remove the old cooler carefully.
- Clean old thermal paste from the CPU and cooler base using isopropyl alcohol and a lint-free cloth.
- Allow both surfaces to dry completely.

3. Apply the thermal paste

- Place a **small pea-sized dot** of thermal paste in the center of the CPU.
- Do not use too much paste; excessive paste can spread beyond the CPU surface.
- For most modern CPUs, the center-dot method is simple and effective.

4. Mount the CPU cooler

- Position the cooler directly over the CPU without sliding it around.
- Lower it straight down onto the CPU.
- Secure the cooler using the manufacturer's mounting screws or clips.

5. Tighten evenly

- If there are multiple screws, tighten them gradually in a **cross/diagonal pattern**.
- Do not overtighten the screws.
- Make sure the cooler is firmly and evenly seated.

6. Connect the fan

- Connect the cooler's fan cable to the motherboard's **CPU_FAN** header.
- Check that the fan cable cannot touch the fan blades.

7. Check your work

- Make sure the cooler does not move or wobble.
- Confirm that the CPU fan is connected.
- Start the PC and monitor CPU temperature using a tool such as HWMonitor or Core Temp.

Common Mistakes to Avoid

Mistake	Why to Avoid It
Using too much thermal paste	Can create unnecessary mess and does not improve cooling
Using too little paste	May leave parts of the CPU surface poorly covered
Reusing old/dried paste	Reduces effective heat transfer
Sliding the cooler around after placing it	Can create an uneven paste layer
Unevenly tightening screws	Can cause uneven contact between CPU and cooler
Forgetting the CPU_FAN connection	Can cause fan/CPU cooling problems

Recommended method: Clean both surfaces → apply a small pea-sized dot → lower the cooler straight down → tighten screws diagonally and evenly → connect **CPU_FAN** → check temperatures.

3. Calculate the recommended PSU wattage for three different PC builds: a basic office PC, a gaming PC, and a content creator PC (with GPU and multiple drives). List all major components and show your calculations for each build.

-> Below is an **assignment-ready example**. The wattages are estimated component power figures; actual consumption varies by hardware model.

1. Basic Office PC

Component	Estimated Power
CPU	65 W
Motherboard	50 W
RAM (16 GB)	10 W
Integrated graphics	15 W
500 GB SSD	5 W
Case fans/USB devices	15 W
Estimated total	160 W

Calculation:

160 W × 1.30 safety margin = **208 W**

Recommended PSU: 300 W

2. Gaming PC

Component	Estimated Power
CPU	100 W
Motherboard	60 W
RAM (32 GB)	15 W
Gaming GPU	250 W
1 TB NVMe SSD	7 W
Case fans	20 W
Other devices	15 W
Estimated total	467 W

Calculation:

467 W × 1.30 safety margin = **607 W**

Recommended PSU: 650 W

3. Content Creator PC

Component	Estimated Power
High-performance CPU	170 W
Motherboard	70 W
RAM (64 GB)	25 W
Dedicated GPU	350 W
2 × NVMe SSD	14 W
2 × HDD	16 W
CPU cooler/pump + fans	35 W
Other USB devices	20 W
Estimated total	700 W

Calculation:

700 W × 1.30 safety margin = **910 W**

Recommended PSU: 1000 W

Summary

Build	Estimated Load	Recommended PSU
Office PC	160 W	300 W
Gaming PC	467 W	650 W
Content Creator PC	700 W	1000 W

Note: For a real build, check the exact CPU/GPU manufacturer's power requirements and PSU recommendations. A quality PSU with appropriate efficiency certification and sufficient headroom is preferable to choosing solely by wattage.

4. Compare the total 5-year electricity cost for two PSUs: one rated 80 Plus Bronze (85% efficiency) and one rated 80 Plus Gold (90% efficiency), both running a 400W average load for 8 hours daily at ₹8 per kWh. Show your math and final cost difference.

Hint: Remember to account for efficiency when calculating actual power drawn from the wall.

-> **Given Data**

- Average PC load = **400 W**
- Usage = **8 hours/day**
- Electricity rate = **₹8/kWh**
- Period = **5 years**
- Bronze efficiency = **85%**
- Gold efficiency = **90%**

1. 80 Plus Bronze (85%)

Actual power drawn from the wall:

$$400 \div 0.85 = 470.59 \text{ W} = 0.47059 \text{ kW}$$

Daily energy:

$$0.47059 \times 8 = 3.7647 \text{ kWh/day}$$

5-year energy:

$$3.7647 \times 365 \times 5 = 6,871.5 \text{ kWh}$$

5-year electricity cost:

$$6,871.5 \times ₹8 = ₹54,972$$

2. 80 Plus Gold (90%)

Actual power drawn from the wall:

$$400 \div 0.90 = 444.44 \text{ W} = 0.44444 \text{ kW}$$

Daily energy:

$$0.44444 \times 8 = 3.5556 \text{ kWh/day}$$

5-year energy:

$$3.5556 \times 365 \times 5 = 6,488.9 \text{ kWh}$$

5-year electricity cost:

$$6,488.9 \times ₹8 = ₹51,911$$

Final Comparison

PSU	Efficiency	5-Year Electricity Cost
80 Plus Bronze	85%	₹54,972
80 Plus Gold	90%	₹51,911
Difference	—	₹3,061 saved with Gold

Conclusion: At a constant 400 W load for 8 hours every day, the **90% efficient Gold PSU** would **save approximately ₹3,061 in electricity costs over 5 years** compared with the 85% efficient Bronze PSU.

5. Create a table comparing air cooling vs liquid cooling for desktop PCs, listing at least 3 pros and 3 cons for each based on performance, noise, maintenance, and cost.

-> Air Cooling vs Liquid Cooling

Factor	Air Cooling	Liquid Cooling (AIO)
Performance – Pros	Simple and reliable cooling; good for most CPUs	Generally better heat dissipation, especially for high-end CPUs
Performance – Cons	Large coolers may struggle with very high CPU loads	Performance depends on radiator size, pump, and airflow
Noise – Pros	No pump noise; can be very quiet with a good fan	Large radiator can allow lower fan speeds under heavy loads
Noise – Cons	Fans may become loud at high CPU temperatures	Has both fan and pump noise
Maintenance – Pros	Very little maintenance; mainly dust cleaning	Sealed AIO systems usually need little routine maintenance

Factor	Air Cooling	Liquid Cooling (AIO)
Maintenance – Cons	Large heatsinks can collect dust	Pump failure or coolant leakage can eventually occur
Cost – Pros	Many effective air coolers are inexpensive	Can provide strong cooling without an extremely large heatsink
Cost – Cons	High-end air coolers can still be expensive	Usually costs more than a comparable air cooler

Summary

Air cooling is a practical choice for most office, gaming, and general-purpose PCs because it is affordable, simple, and reliable. **Liquid cooling** is useful when a system has a high-performance CPU or sustained heavy workloads where additional cooling capacity is beneficial.